

Chi-Square Analysis

Stroke Status & Hospital Readmission Rates

Research Question

Do patients that come into the hospital due to a stroke have a higher rate of readmission than the overall patient population?

Stakeholder Benefit

If patients who experience a stroke are being readmitted more often than average, the hospital should adjust by asking stroke patients to stay longer for their safety.

Methodology

The chi square was chosen as this test will reveal the strength of the relationship between two categorical variables, in this case, stroke and readmission. The null hypothesis can be put to test, revealing the independence of the two chosen categorical variables. The categorical readmission and stroke columns will be the required data for this chi square analysis.

Analysis Output

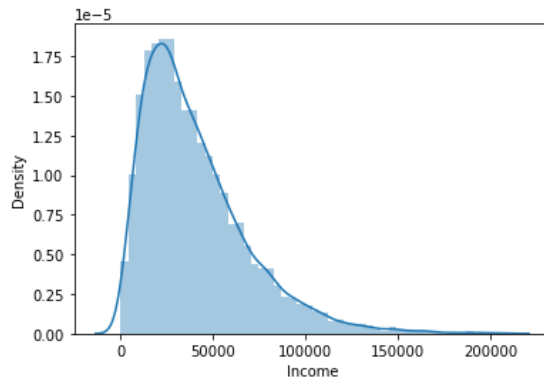
```
In [36]: runfile('C:/Users/velez/OneDrive/Desktop/School/D207/Brandon Velez D207 PA.py', wdir='C:/Users/velez/OneDrive/Desktop/School/D207')
CaseOrder Customer_id ... Item7 Item8
0          1      C412403 ...      3      4
1          2      Z919181 ...      3      3
2          3      F995323 ...      3      3
3          4      A879973 ...      5      5
4          5      C544523 ...      4      3

[5 rows x 50 columns]
Stroke      No      Yes
ReAdmis
No          5071    1260
Yes         2936     733
Stroke      No      Yes
ReAdmis
No          0.800979  0.199021
Yes         0.800218  0.199782
0.9474770077616069
p value is 0.9474770077616069
Independent (accept H0)
```

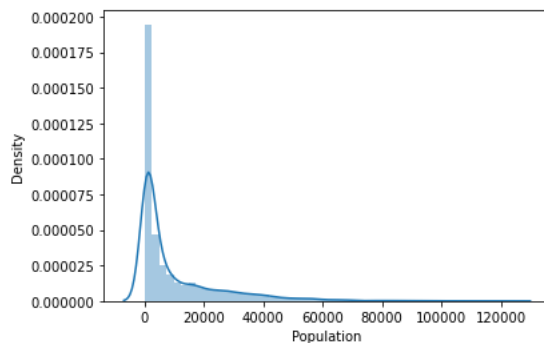
Univariate Analysis

The income density graph is right skewed, meaning there is a high peak in the lower values on the left, and then has a very long tail as values increase to the right.

The population density graph is heavily right skewed, as there is an extremely high peak in density early on, and then tails off for quite a long time as the population values increase and density decreases.

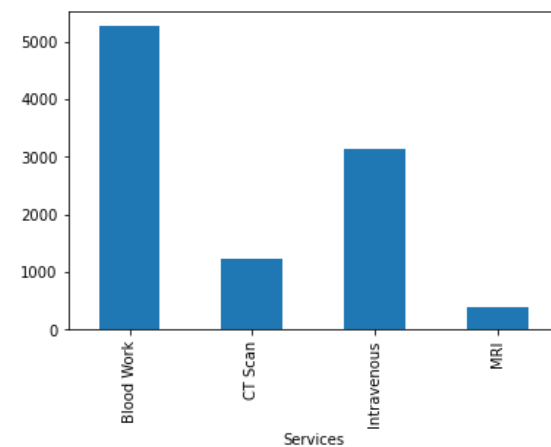
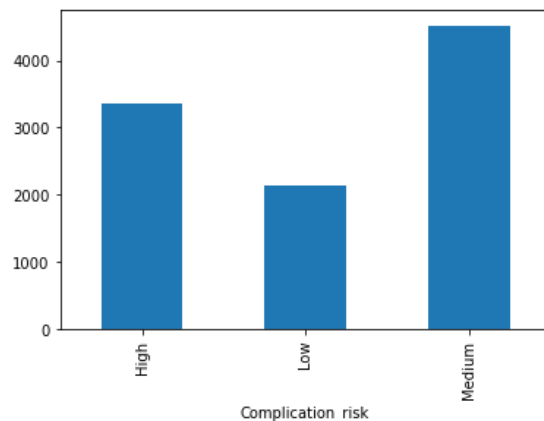


```
sns.distplot(df.Income)
Out[4]:
count    10000.000000
mean      40490.495160
std       28521.153293
min        154.080000
25%       19598.775000
50%       33768.420000
75%       54296.402500
max       207249.100000
Name: Income, dtype: float64
```



```
sns.distplot(df.Population)
Out[5]:
count    10000.000000
mean       9965.253800
std       14824.758614
min         0.000000
25%        694.750000
50%       2769.000000
75%      13945.000000
max      122814.000000
Name: Population, dtype: float64
```

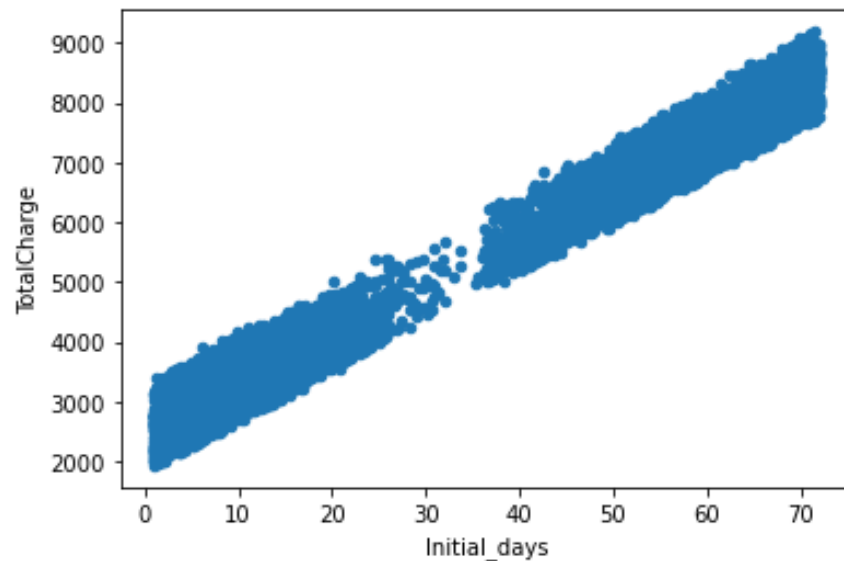
The categorical complication risk distribution chart is a discrete non-uniform distribution. The categorical services chart follows a non-uniform distribution as well.



Bivariate Analysis

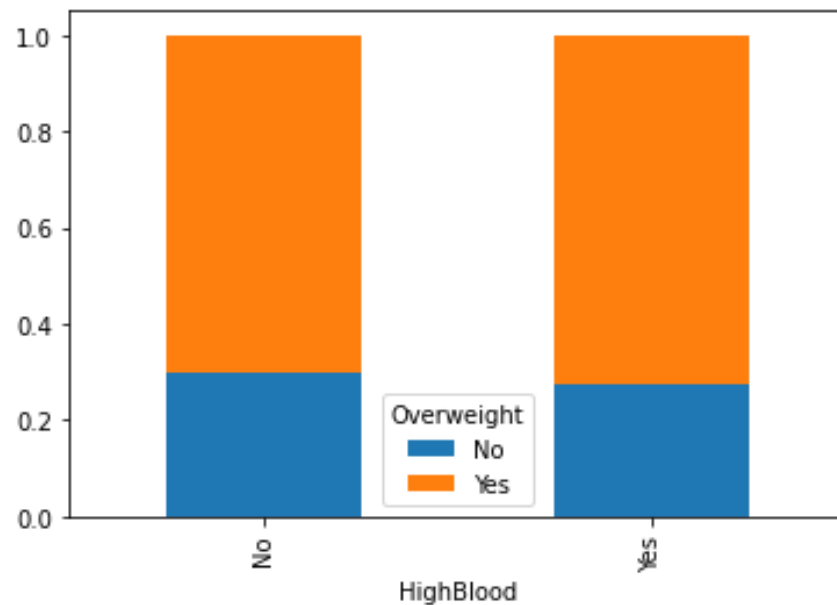
The initial days to total charge visualization follows a linear distribution, meaning more days will correlate very strongly to a higher total charge. This is not perfect however, as any value of initial days can correlate to multiple total charge values.

Overweight to high blood pressure is a discrete uniform distribution, meaning there is no significant difference in blood pressure classification when comparing overweight to non overweight patients.



```
Out[6]:  
count    10000.000000  
mean      5312.172769  
std       2180.393838  
min       1938.312067  
25%       3179.374015  
50%       5213.952000  
75%       7459.699750  
max       9180.728000  
Name: TotalCharge, dtype: float64
```

```
Out[8]:  
count    10000.000000  
mean       34.455299  
std        26.309341  
min         1.001981  
25%         7.896215  
50%        35.836244  
75%        61.161020  
max        71.981490  
Name: Initial_days, dtype: float64
```



Results

The Null hypothesis was proven to be true, as stroke patients do not have a statistically significant difference in readmission rate compared to the overall patient population. Based on the results of the hypothesis test, no action is needed on the hospital's part to adjust how they are handling patients who have experienced a stroke. There is no statistical indication of a higher readmission rate for stroke victims or any other clear issue.

Limitations

A limitation of this data analysis would be that this data is only coming from a singular hospital location over a short period of time. While these results are valid in the context of the provided data, this is too narrow in scope to make any general assumptions on the relationship between stroke status and readmission rate. If thousands of hospitals were sampled across the world over a period of multiple decades, the data analysis would be resistant to any outlying values or geographical differences, and could therefore be generalized and considered statistically robust.

There will always be the issue of self-reporting bias, as patients could be incorrect, write illegibly on intake forms, or just lie. The hospital provided fields are subject to human error, but likely much less so than patient reported information as trained staff are inputting that data.